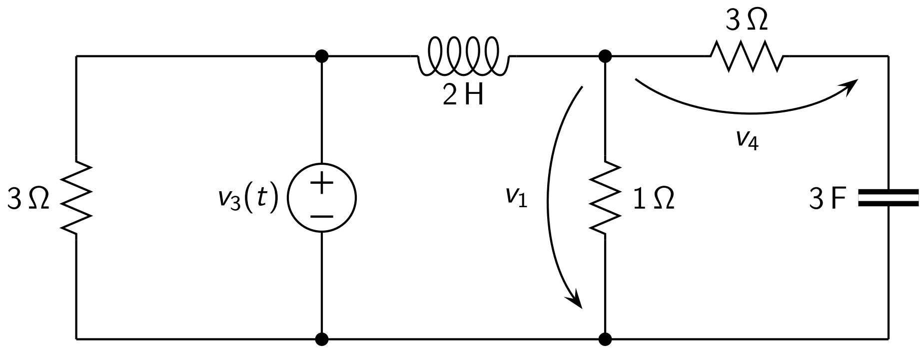
Problem: the circuit operates at steady-state. Find the expression of $v_1(t)$, $v_4(t)$.
Data: $v_3(t) = 3\cos(3t + 165^o)\,\mathrm{V}$.



Solution

$$v_1(t) = 0.373\cos(3\,t - 98^o)\,\mathrm{V}$$

$$v_4(t) = 0.372\cos(3\,t - 96^o)\,\mathrm{V}$$